

— NETWORK & SECURITY ENGINEER

# Hassan Ismail.

CS & AI Student, South Valley National University

Working across two tracks: competitive programming in C++ (ECPC/ICPC) and network security. On the security side, I design and harden network infrastructure — firewall clusters, SD-WAN edges, site-to-site IPsec tunnels, and segmented DMZ architectures on FortiGate — then prove every path with a policy trace and a CLI ping.

“Failure is a milestone for success.”

Hassanismailtheemperor@gmail.com

+20 103 270 2926

LinkedIn Profile

Qena, Egypt

## SELECTED WORK

### 01 HA Cluster <-> SD-WAN Site-to-Site VPN

Two FortiGate sites joined over an encrypted tunnel

### 02 DMZ Segmentation

Three-legged firewall isolating public-facing services



ID: H. ISMAIL

● AVAILABLE

# What's actually behind the work

Every item here is exercised somewhere in the two builds in this book.

## CORE SKILLS

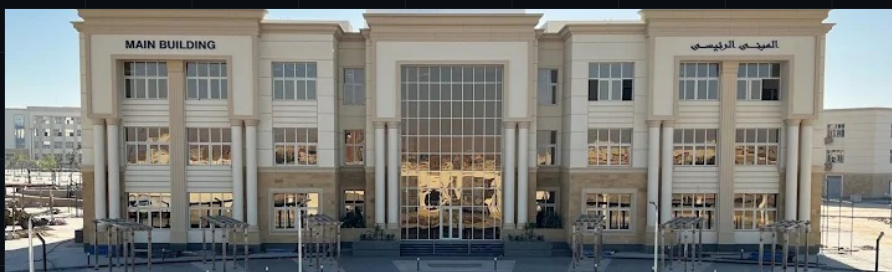
- Firewall Policy Design
- High Availability (HA) Clustering
- Site-to-Site IPsec VPN
- SD-WAN (Multi-WAN)
- Network Segmentation (DMZ)
- Static & Failsafe Routing
- CLI-Based Verification & Diagnostics

## TOOLS & PLATFORMS

- FortiOS / FortiGate VM64
- FortiGate CLI Console
- Python (Intro + Intermediate)
- Network Fundamentals & Protocols

# Currently building the fundamentals

Second-year Computer Science & AI student, with applied network-security practice and two completed Python courses alongside coursework.



## ENTERING YEAR 3

### South Valley National University

Faculty of Computer Science & Artificial Intelligence - CS & IT

Completed the second year, now entering the third. Coursework in programming, algorithms, and computer networks runs alongside the independent FortiGate work in this book.



## NTI - 2026

### National Telecommunication Institute

FortiOS Administrator -- Network Security, 60-hour track

Official Fortinet Academy curriculum: firewall policies, the Security Fabric, authentication, and security profiles (IPS, antivirus, web filtering, app control), plus IPsec VPN, SD-WAN, and HA -- the practical basis for the FortiGate projects in this book, built toward the NSE4 FortiOS Administrator path (exam NSE4\_FGT\_AD-7.6). Also completed two Microsoft-backed Python courses under an MCIT digital-skills program.



#### Intro to Python

Sep 14-23, 2025



#### Python - Intermediate

Sep 25-Oct 2, 2025

# C++ contest record

Alongside the security work in this book, I compete in C++ with South Valley National University's ECPC/ICPC team.

APR 2025 - INDIVIDUAL

**ICPC ACPC Kickoff  
Online Individual Contest**  
**Honorable Mention**



AUG 2025 - INDIVIDUAL

**ECPC Qualifications  
Collegiate Prog. Contest, Day 8**  
**94th Place**



AUG 2025 - TEAM

**ECPC Qualifications  
Collegiate Prog. Contest, Day 8**  
**94th Place**



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## ● COMPETITIVE PROGRAMMING -- CONTINUED

AUG 2026 - INDIVIDUAL

**ECPC Qualifications  
Collegiate Prog. Contest, Day 8**  
**93rd Place**



AUG 2026 - TEAM

**ECPC Qualifications  
Collegiate Prog. Contest, Day 8**  
**93rd Place**



# Services

Scope drawn directly from the two builds in this book.

01

## Firewall & Policy Design

Rule sets built around least privilege - explicit deny where it matters, scoped ACCEPT rules everywhere else, all under UTM inspection.

Project 01 - 02

02

## High Availability

Active-passive FortiGate clustering with unit priority and synchronization, so a single hardware failure doesn't take the network down.

Project 01

03

## Site-to-Site VPN

IPsec tunnels between independent sites, with routing that prefers the tunnel and falls back to a blackhole route if it drops.

Project 01

04

## SD-WAN

Multi-WAN edges grouped into a single zone, so outbound traffic keeps moving even if one link degrades.

Project 01

05

## Network Segmentation

Zone-based design - LAN, DMZ, WAN - with firewall policy controlling what can reach what.

Project 02

06

## Verification & Testing

Every path is confirmed from the CLI - sourced pings, tunnel status, session counters - before it's called done.

Project 01 - 02

— PROJECT 01

# HA Cluster <-> SD-WAN Site-to-Site VPN

Two independent FortiGate deployments, built and validated as a single connected system: an active-passive HA pair at Site P1, and a dual-WAN SD-WAN edge at Site P2, joined by an IPsec tunnel and reachable subnet-to-subnet. Static routes on both sides prefer the tunnel and fall back to a blackhole route so traffic never leaks to the internet if the VPN drops.

Role: Primary / Secondary

Priority: 250

Tunnel: Up - both directions

Verification: CLI ping, 0% loss

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Administrators

Admin Profiles

Firmware

Fabric Management

Settings

HA

SNMP

Replacement Messages

FortiGuard

Feature Visibility

Certificates

Security Fabric

Log & Report

HA: Primary

admin

FortiGate VM64

FortiGate VM64

FortiGate-VM64-HA-backup (Primary)

FortiGate-VM64 (Secondary)

Refresh

Edit

Remove device from HA cluster

| Status       | Priority | Hostname         | Serial No.       | Role      | System Uptime | Sessions | Throughput |
|--------------|----------|------------------|------------------|-----------|---------------|----------|------------|
| Synchronized | 250      | FortiGate-VM64-H | FGVMEVJZDR22AM0A | Primary   | 1m 15s        | 42       | 78.00 kbps |
| Synchronized | 250      | FortiGate-VM64   | FGVMEVJWGIS_7T31 | Secondary | 3m 32s        | 27       | 22.00 kbps |

FortiNET

v7.0.15

2

01 HA Status

Active-passive HA pair, both synchronized at priority 250 - live session count and throughput per unit.



Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New

Edit

Delete

Integrate Interface

Search

Group By Type

| Name                  | Type               | Members | IP/Netmask                   | Administrative Access                      | DHCP Clients | DHCP Ranges             | Ref |
|-----------------------|--------------------|---------|------------------------------|--------------------------------------------|--------------|-------------------------|-----|
| 802.3ad Aggregate 1   |                    |         |                              |                                            |              |                         |     |
| fortilink             | 802.3ad Aggregate  |         | Dedicated to FortiSwitch     | PING<br>Security Fabric Connection         |              | 10.255.1.2-10.255.1.254 | 2   |
| Physical Interface 11 |                    |         |                              |                                            |              |                         |     |
| port1                 | Physical Interface |         | 172.21.111.113/255.255.240.0 | PING<br>HTTPS<br>SSH<br>HTTP<br>FMG-Access |              |                         | 4   |
| port2                 | Physical Interface |         | 0.0.0.0/0.0.0.0              |                                            |              |                         | 0   |
| port3                 | Physical Interface |         | 192.168.10.1/255.255.255.0   | PING<br>HTTPS<br>SSH                       |              |                         | 3   |
| port4                 | Physical Interface |         | 0.0.0.0/0.0.0.0              |                                            |              |                         | 0   |

0 Security Rating Issues

0% 13 Updated: 09:36:19

02 Interfaces

port1 carries WAN/management (SSH, HTTPS, FMG-Access); port3 terminates the internal 192.168.10.0/24 LAN.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

- Interfaces
- DNS
- Packet Capture
- SD-WAN
- Static Routes
- Policy Routes
- RIP
- OSPF
- BGP
- Routing Objects
- Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

HA: Primary

Create NewEditCloneDeleteSearch

| Destination      | Gateway IP                    | Interface | Status  | Comments                               |
|------------------|-------------------------------|-----------|---------|----------------------------------------|
| 0.0.0.0/0        | Dynamic Gateway (172.21.96.1) | port1     | Enabled |                                        |
| vpn to p2_remote |                               | vpn to p2 | Enabled | VPN: vpn to p2 (Created by VPN wizard) |
| vpn to p2_remote |                               | Blackhole | Enabled | VPN: vpn to p2 (Created by VPN wizard) |

3

03 Static Routing

Default route out port1, a primary route through the tunnel, and a blackhole route so nothing leaks if the VPN drops.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

HA: Primary

Create New

Edit

Delete

Policy Lookup

Search

Export

Interface Pair View

By Sequence

| Name                | Source | Destination | Schedule | Service | Action | NAT | Security Profiles | Log | Bytes |
|---------------------|--------|-------------|----------|---------|--------|-----|-------------------|-----|-------|
| port3 → port1 1     |        |             |          |         |        |     |                   |     |       |
| LAN to internet     | all    | all         | always   | ALL     | ACCEPT |     | SSL no-inspection | UTM | 0 B   |
| port3 → vpn to p2 1 |        |             |          |         |        |     |                   |     |       |
| vpn to p2 → port3 1 |        |             |          |         |        |     |                   |     |       |
| Implicit 1          |        |             |          |         |        |     |                   |     |       |

0 Security Rating Issues

4 Updated: 09:37:18

04 Firewall Policy

LAN-to-internet plus dedicated inbound/outbound rules for the VPN interface, all under UTM inspection.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

HA: Primary

>

🔍

🔔 2

admin

+ Create New

Edit

Delete

Search

| Tunnel                     | Interface Binding | Status | Ref. |
|----------------------------|-------------------|--------|------|
| Site to Site - FortiGate 1 |                   |        |      |
| vpn to p2                  | port1             | Up     | 4    |

FORTINET v7.0.15

0 Security Rating Issues

1

05 IPsec Tunnel

Site-to-site tunnel "vpn to p2," bound to port1, status Up.

## ● HASSAN ISMAIL

### ● SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

The screenshot shows the FortiGate VM64-HA-backup CLI console. The left sidebar contains navigation options: Dashboard, Network, Policy & Objects, Security Profiles, VPN, Overlay Controller, IPsec Tunnels, IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System, Security Fabric, and Log & Report. The main console area displays the following commands and output:

```
FortiGate-VM64-HA-backup # execute ping-options source 192.168.10.1
FortiGate-VM64-HA-backup # execute ping 192.168.20.1
PING 192.168.20.1 (192.168.20.1): 56 data bytes
64 bytes from 192.168.20.1: icmp_seq=0 ttl=255 time=2.4 ms
64 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=1.2 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=1.7 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=1.5 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=1.6 ms

--- 192.168.20.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.2/1.6/2.4 ms

FortiGate-VM64-HA-backup #
```

The bottom status bar shows the Fortinet logo, version v7.0.15, and a security rating of 0. The user 'admin' is logged in.

### 06 CLI Verification

Sourced ping from the P1 LAN to the P2 LAN (192.168.20.1) - 5/5 packets, 0% loss, 1.6 ms average.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

The screenshot shows the FortiGate VM64 SD-WAN configuration interface. The left sidebar contains a navigation menu with options like Dashboard, Network, Interfaces, DNS, Packet Capture, SD-WAN, Static Routes, Policy Routes, RIP, OSPF, BGP, Routing Objects, Multicast, Policy & Objects, Security Profiles, VPN, User & Authentication, System, Security Fabric, and Log & Report. The main content area displays a table of interfaces. At the top, there is a 'FortiGate VM64' section with a grid of 24 small icons representing different interfaces. Below this, there is a 'Create New' button, an 'Edit' button, a 'Delete' button, and an 'Integrate Interface' button. A search bar and a 'Group By Type' dropdown are also present. The table has columns for Name, Type, Members, IP/Netmask, Administrative Access, DHCP Clients, DHCP Ranges, and Ref. The table lists four physical interfaces: port1, port2, port3, and port4. port1 and port2 are configured with IP addresses 172.21.109.73/255.255.240.0 and 172.21.98.128/255.255.240.0 respectively. port3 is configured with IP address 192.168.20.1/255.255.255.0. port4 is configured with IP address 0.0.0.0/0.0.0.0. The table also shows administrative access for each interface, including PING, HTTPS, SSH, HTTP, and FMG-Access. The bottom of the interface shows a 'Security Rating Issues' section with 0 issues and a '0% 14' status. The bottom right corner shows the date and time 'Updated: 09:42:43' and a refresh button.

| Name                         | Type               | Members | IP/Netmask                  | Administrative Access                      | DHCP Clients | DHCP Ranges | Ref. |
|------------------------------|--------------------|---------|-----------------------------|--------------------------------------------|--------------|-------------|------|
| <b>802.3ad Aggregate 1</b>   |                    |         |                             |                                            |              |             |      |
| fortilink                    | 802.3ad Aggregate  |         | Dedicated to FortiSwitch    | PING<br>Security Fabric Connection         |              |             | 1    |
| <b>Physical Interface 11</b> |                    |         |                             |                                            |              |             |      |
| port1                        | Physical Interface |         | 172.21.109.73/255.255.240.0 | PING<br>HTTPS<br>SSH<br>HTTP<br>FMG-Access |              |             | 3    |
| port2                        | Physical Interface |         | 172.21.98.128/255.255.240.0 | PING<br>HTTPS<br>SSH                       |              |             | 1    |
| port3                        | Physical Interface |         | 192.168.20.1/255.255.255.0  | PING<br>HTTPS<br>SSH                       |              |             | 3    |
| port4                        | Physical Interface |         | 0.0.0.0/0.0.0.0             |                                            |              |             | 0    |

0 Security Rating Issues 0% 14 Updated: 09:42:43

## 01 Interfaces

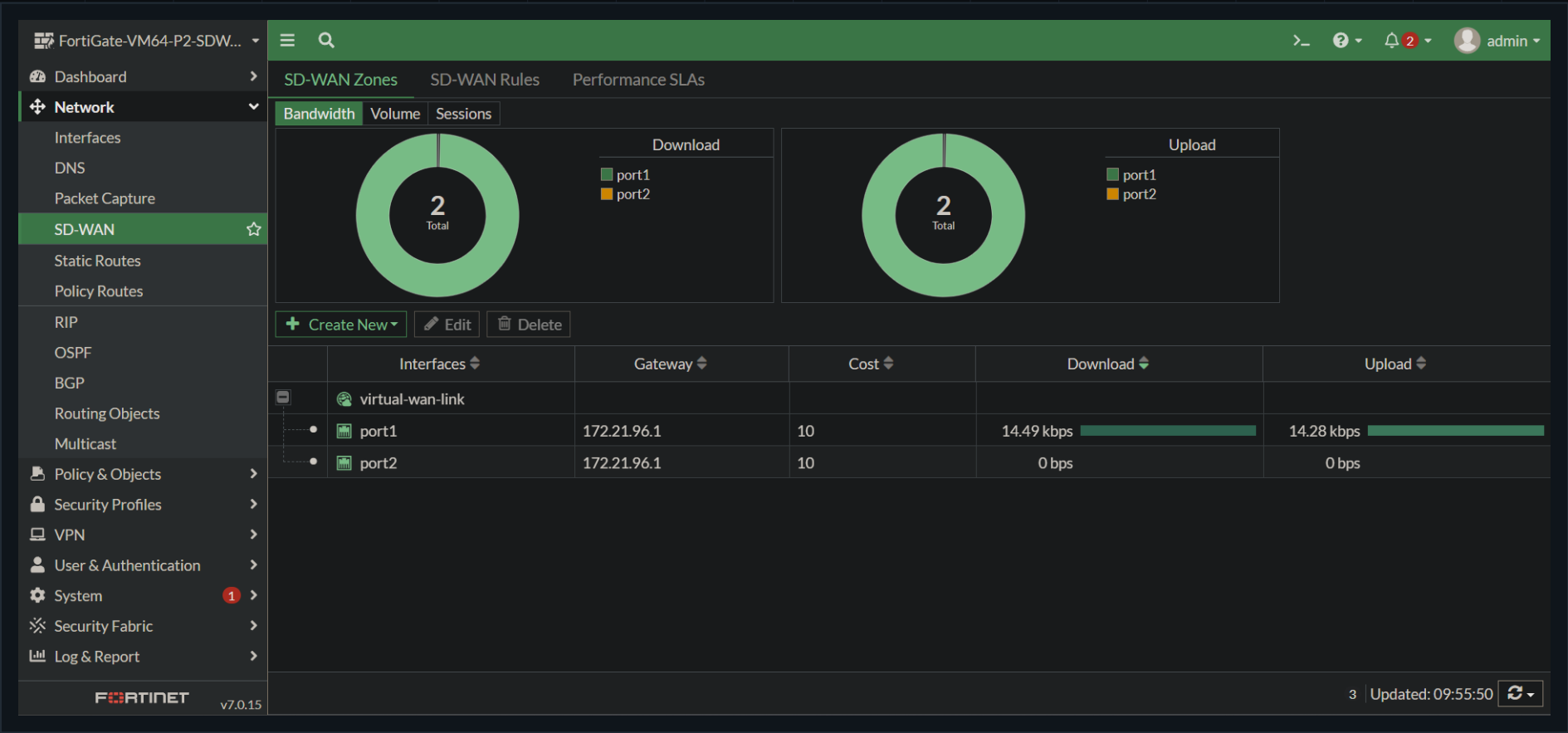
Two independent WAN links (port1, port2) plus the internal 192.168.20.0/24 LAN on port3.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up



02 SD-WAN Zone

Both WAN ports grouped into the virtual-wan-link at equal cost, with live download/upload throughput per member.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...

Dashboard

Network

- Interfaces
- DNS
- Packet Capture
- SD-WAN
- Static Routes
- Policy Routes
- RIP
- OSPF
- BGP
- Routing Objects
- Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FORTINET v7.0.15

Create New Edit Clone Delete Search

| Destination      | Gateway IP | Interface        | Status  | Comments                               |
|------------------|------------|------------------|---------|----------------------------------------|
| 0.0.0.0/0        |            | virtual-wan-link | Enabled |                                        |
| vpn to p1_remote |            | vpn to p1        | Enabled | VPN: vpn to p1 (Created by VPN wizard) |
| vpn to p1_remote |            | Blackhole        | Enabled | VPN: vpn to p1 (Created by VPN wizard) |

3

03 Static Routing

Default route rides the SD-WAN zone; the route to P1 gets the same primary-plus-blackhole failsafe pattern as P1.



WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...
Dashboard
Network
Policy & Objects
Firewall Policy
IPv4 DoS Policy
Addresses
Internet Service Database
Services
Schedules
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Traffic Shaping
Security Profiles
VPN
User & Authentication
System
Security Fabric
Log & Report

Create New
Edit
Delete
Policy Lookup
Search
Export
Interface Pair View
By Sequence

| Name                       | Source | Destination | Schedule | Service | Action | NAT     | Security Profiles | Log | Bytes |
|----------------------------|--------|-------------|----------|---------|--------|---------|-------------------|-----|-------|
| port3 → virtual-wan-link 1 |        |             |          |         |        |         |                   |     |       |
| lan to sdwan               | all    | all         | always   | ALL     | ACCEPT | Enabled | SSL no-inspection | UTM | 0 B   |
| port3 → vpn to p1 1        |        |             |          |         |        |         |                   |     |       |
| vpn to p1 → port3 1        |        |             |          |         |        |         |                   |     |       |
| Implicit 1                 |        |             |          |         |        |         |                   |     |       |

0 Security Rating Issues
4 Updated: 09:43:09

## 04 Firewall Policy

LAN-to-SD-WAN policy for general egress, mirrored by inbound/outbound rules for the tunnel back to P1.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...

Dashboard

Network

Policy & Objects

Security Profiles

VPN

- Overlay Controller VPN
- IPsec Tunnels
- IPsec Wizard
- IPsec Tunnel Template
- SSL-VPN Portals
- SSL-VPN Settings
- SSL-VPN Clients
- VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

Create New

Edit

Delete

Search

| Tunnel                     | Interface Binding | Status | Ref. |
|----------------------------|-------------------|--------|------|
| Site to Site - FortiGate 1 |                   |        |      |
| vpn to p1                  | port1             | Up     | 4    |

0 Security Rating Issues

1

05 IPsec Tunnel

Site-to-site tunnel "vpn to p1," bound to port1, status Up.

## ● HASSAN ISMAIL

### ● SITE P2 -- SD-WAN

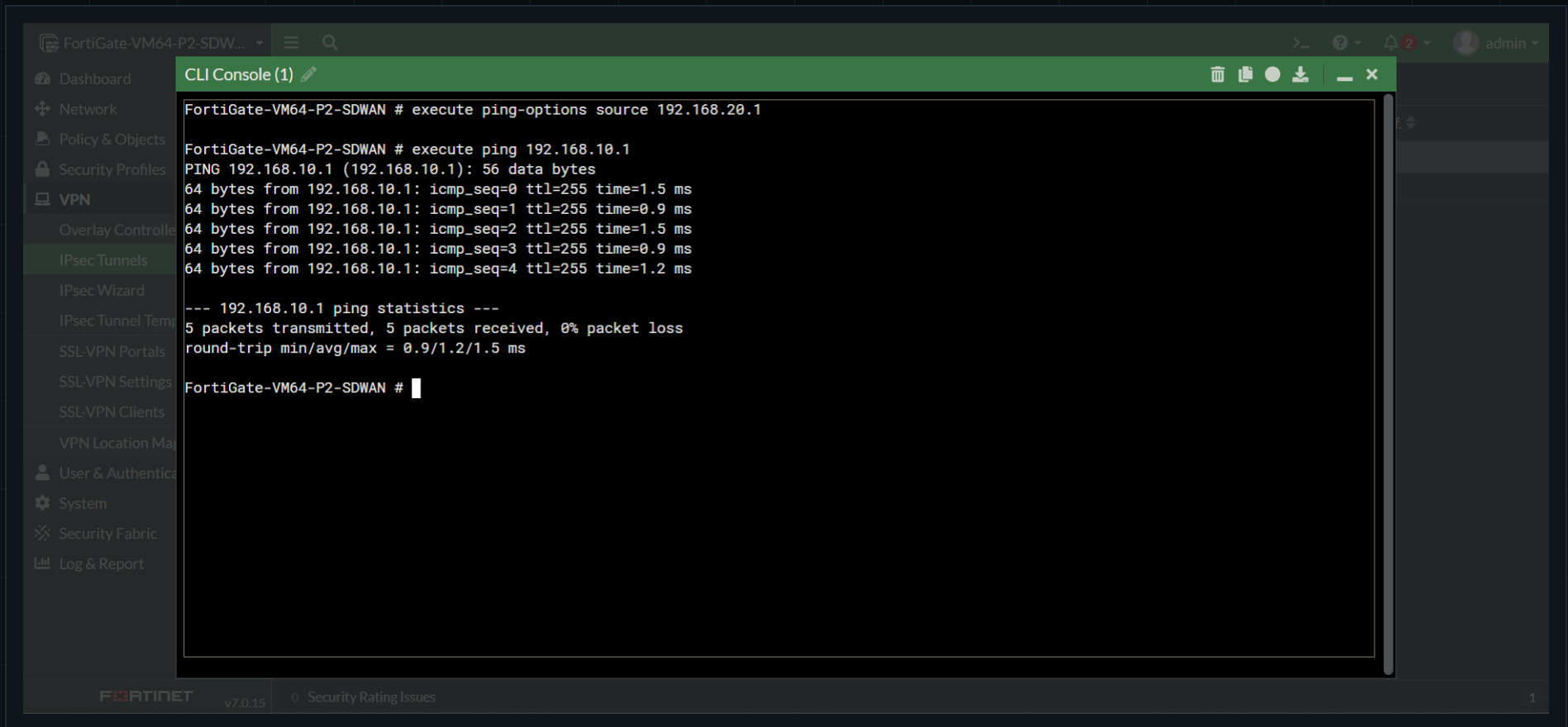
PROJECT 01

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up



The screenshot shows the FortiGate VM64-P2-SDWAN CLI console. The left sidebar contains a navigation menu with options: Dashboard, Network, Policy & Objects, Security Profiles, VPN, Overlay Controller, IPsec Tunnels, IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System, Security Fabric, and Log & Report. The main console area displays the following output:

```
FortiGate-VM64-P2-SDWAN # execute ping-options source 192.168.20.1

FortiGate-VM64-P2-SDWAN # execute ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1): 56 data bytes
64 bytes from 192.168.10.1: icmp_seq=0 ttl=255 time=1.5 ms
64 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=0.9 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=1.5 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=0.9 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=1.2 ms

--- 192.168.10.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.9/1.2/1.5 ms

FortiGate-VM64-P2-SDWAN #
```

The bottom status bar shows the Fortinet logo, version v7.0.15, and a security rating of 0.

## 06 CLI Verification

Sourced ping from the P2 LAN to the P1 LAN (192.168.10.1) - 5/5 packets, 0% loss, 1.2 ms average, confirming bidirectional reachability.

— PROJECT 02

# DMZ Segmentation

A single FortiGate split into three security zones, so a compromised public-facing host in the DMZ can never pivot into the internal LAN. port1 faces the WAN, port2 hosts the internal LAN (192.168.10.0/24), and port3 is a dedicated DMZ (192.168.20.0/24) for exposed services. Policy does the isolating: DMZ-to-LAN traffic is explicitly denied, LAN-to-DMZ and LAN-to-WAN are permitted, and the implicit deny rule closes everything not named.

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

LAN -> WAN: Accepted

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

FortiGate-VM64

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

| Name                  | Type               | Members | IP/Netmask                   | Administrative Access                      | DHCP Clients | DHCP Ranges | Ref. |
|-----------------------|--------------------|---------|------------------------------|--------------------------------------------|--------------|-------------|------|
| 802.3ad Aggregate 1   |                    |         |                              |                                            |              |             |      |
| fortilink             | 802.3ad Aggregate  |         | Dedicated to FortiSwitch     | PING<br>Security Fabric Connection         |              |             | 1    |
| Physical Interface 10 |                    |         |                              |                                            |              |             |      |
| DMZ zone (port3)      | Physical Interface |         | 192.168.20.1/255.255.255.0   | PING<br>HTTPS<br>SSH                       |              |             | 3    |
| LAN network (port2)   | Physical Interface |         | 192.168.10.1/255.255.255.0   | PING<br>HTTPS<br>SSH                       |              |             | 4    |
| port1                 | Physical Interface |         | 172.28.141.165/255.255.240.0 | PING<br>HTTPS<br>SSH<br>HTTP<br>FMG-Access |              |             | 1    |

0 SecurityRating Issues

0% 12 Updated: 09:07:46

01 Interfaces

Dedicated DMZ (port3), internal LAN (port2), and WAN (port1) on a single FortiGate - each its own subnet.

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

FortiGate-VM64
Dashboard
Network
Policy & Objects
Firewall Policy
IPv4 DoS Policy
Addresses
Internet Service Database
Services
Schedules
Virtual IPs
IP Pools
Protocol Options
Traffic Shaping
Security Profiles
VPN
User & Authentication
System
Security Fabric
Log & Report
FORTINET v7.0.15

Create New
Edit
Delete
Policy Lookup
Search
Export
Interface Pair View
By Sequence

| Name                                     | Source      | Destination | Schedule | Service | Action | NAT      | Security Profiles | Log     | Bytes |
|------------------------------------------|-------------|-------------|----------|---------|--------|----------|-------------------|---------|-------|
| DMZ zone (port3) → LAN network (port2) 1 |             |             |          |         |        |          |                   |         |       |
| block dmz to lan                         | dmz network | lan network | always   | ALL     | DENY   |          |                   | All     | 0 B   |
| LAN network (port2) → DMZ zone (port3) 1 |             |             |          |         |        |          |                   |         |       |
| lan to dmz access                        | lan network | dmz network | always   | ALL     | ACCEPT | Disabled | SSL no-inspection | All     | 0 B   |
| LAN network (port2) → port1 1            |             |             |          |         |        |          |                   |         |       |
| lan to wan                               | lan network | all         | always   | ALL     | ACCEPT | Enabled  | SSL no-inspection | All     | 0 B   |
| Implicit 1                               |             |             |          |         |        |          |                   |         |       |
| Implicit Deny                            | all         | all         | always   | ALL     | DENY   |          |                   | Enabled | 0 B   |

0 SecurityRating Issues
4 Updated: 09:07:56

## 02 Firewall Policy

DMZ-to-LAN explicitly denied, LAN-to-DMZ and LAN-to-WAN permitted, implicit deny closes everything else.

● HASSAN ISMAIL

---

— GET IN TOUCH

# Open to network & security engineering work.

Hassanismailtheemperor@gmail.com

+20 103 270 2926

LinkedIn Profile

Qena, Egypt